

Building a Water-Quality Data Platform for Future Health Outcome Research

Analyzing Water Quality in Counties Across Texas

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WHY WATER MATTERS

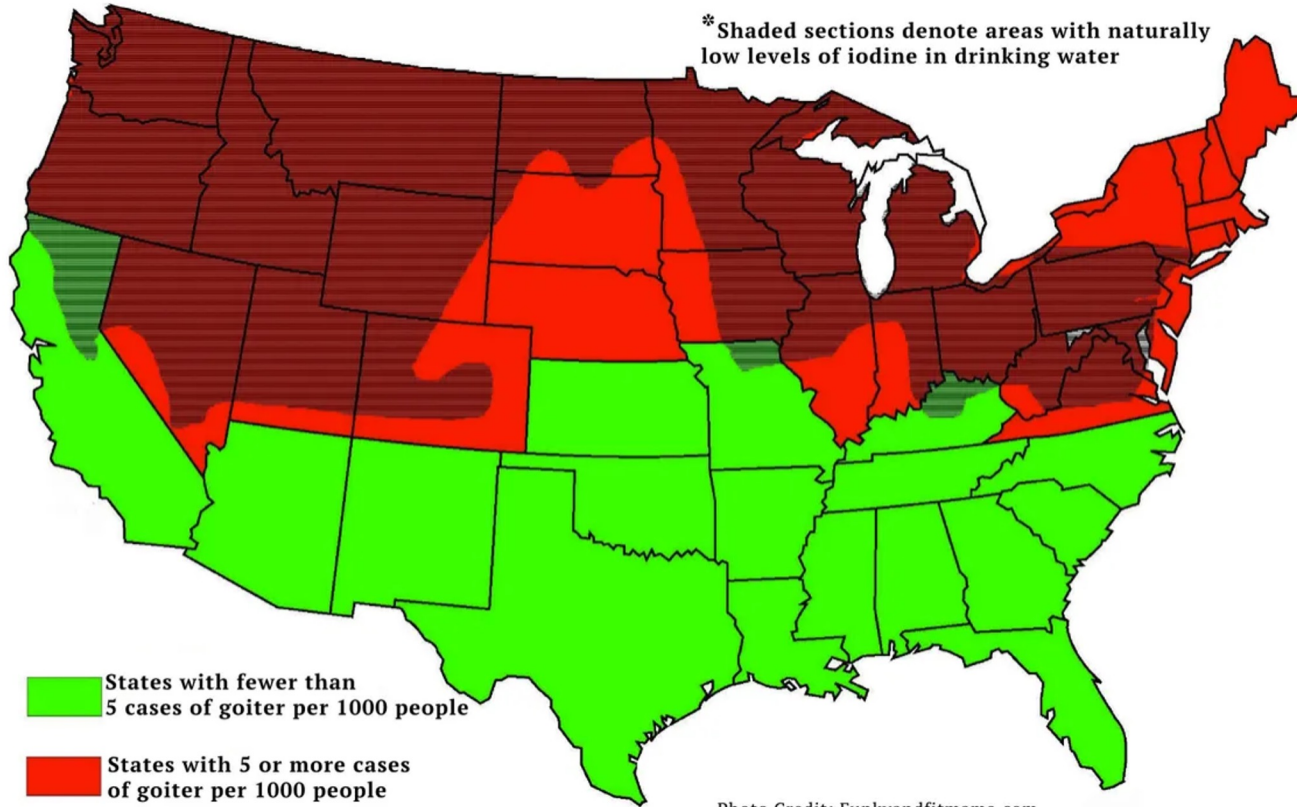


Photo Credit: Funkyandfitmama.com



WATER'S IMPACT

- Arsenic (As) → cancer, heart failure
- Fluoride (F^-) → dental protection at low levels, toxicity at high
- Nitrate (NO_3^-) / Nitrite (NO_2^-) → thyroid, cancer, cognitive effects
- Lithium (Li) → Alzheimer's & Dementia, ADHD & Cognitive Impairment, Depression & Anxiety, Schizophrenia

Texas Water Development Board (TWDB)

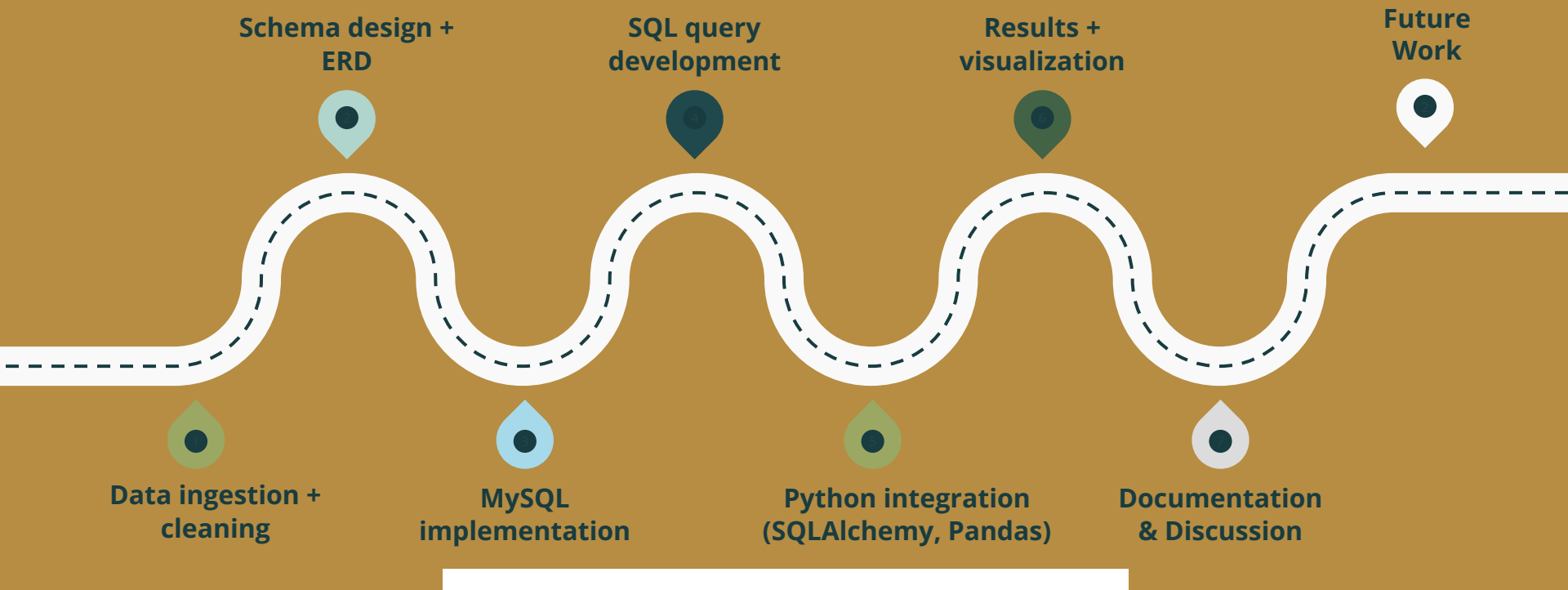
- TWDB has been collecting
 - Water Quality Data
 - For hundreds of minerals and contaminants
 - For every county in Texas
 - For more than 20 years



PROBLEM

- TWDB database is not optimized for questions like:
 - Which counties have the highest arsenic?
 - Do certain minerals co-occur?
 - Are there spatial patterns across Texas?
- Without easy access to the full database, analysis is slow, manual, and inconsistent

PROJECT ROADMAP



PROJECT GOALS

- Collect and clean groundwater quality data
- Build a relational database that supports fast and flexible, analysis at the county level that is 3rd normal form
- Build ERD matching real-world database structure
- Enable EDA, modeling, geo-mapping, integration utilizing various python packages
- Run SQL queries across wells, counties, and minerals to visualize their distributions
- Final Jupyter Notebook deliverable to aid recommendations

DATA OVERVIEW

TWDB DATABASE

- Well data stored in text files
- Well data – State Well ID, locations, counties, aquifers, *inter-alia*
- About 488 components (e.g. arsenic, lithium, nitrate, fluoride, etc.)
- Sampling dates, units of measure, reliability flags, agencies

DATABASE DESIGN SUMMARY

- 64 Database tables
- (3NF) normalized schema
- Primary/foreign keys ensure relational integrity
- Supports joining:
well → sample → parameter → county

COMPONENT ANALYSIS

Analysis Workflow

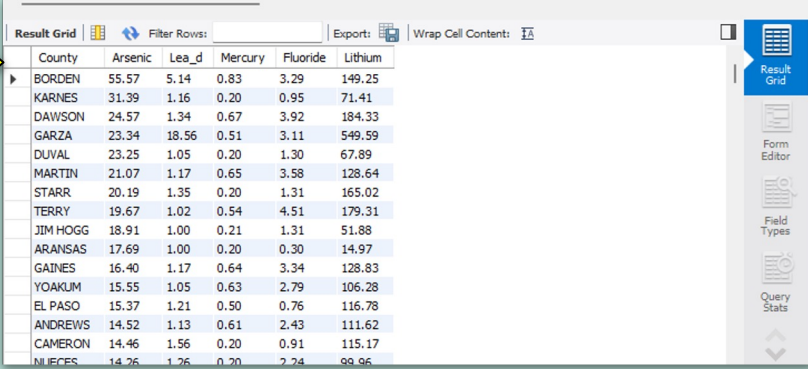
- Query parameter values from `waterqualitydetail` table →
- Join to wells, sample headers, counties →
- Filter by mineral of interest (arsenic, lithium, magnesium, lead etc.) →
- Aggregate functions to county level (AVG, MIN, MAX, COUNT) →
- Prepare tables for Python and mapping

SQL query: Average concentration per mineral per county (2000–2024)

```
SELECT
    c.CountyName AS County,
    ROUND(AVG(CASE WHEN LOWER(wqp.ConstituentName) = 'arsenic' THEN wqd.ParameterValue END), 2) AS Arsenic,
    ROUND(AVG(CASE WHEN LOWER(wqp.ConstituentName) = 'lead' THEN wqd.ParameterValue END), 2) AS Lea_d,
    ROUND(AVG(CASE WHEN LOWER(wqp.ConstituentName) = 'mercury' THEN wqd.ParameterValue END), 2) AS Mercury,
    ROUND(AVG(CASE WHEN LOWER(wqp.ConstituentName) = 'fluoride' THEN wqd.ParameterValue END), 2) AS Fluoride,
    ROUND(AVG(CASE WHEN LOWER(wqp.ConstituentName) = 'lithium' THEN wqd.ParameterValue END), 2) AS Lithium

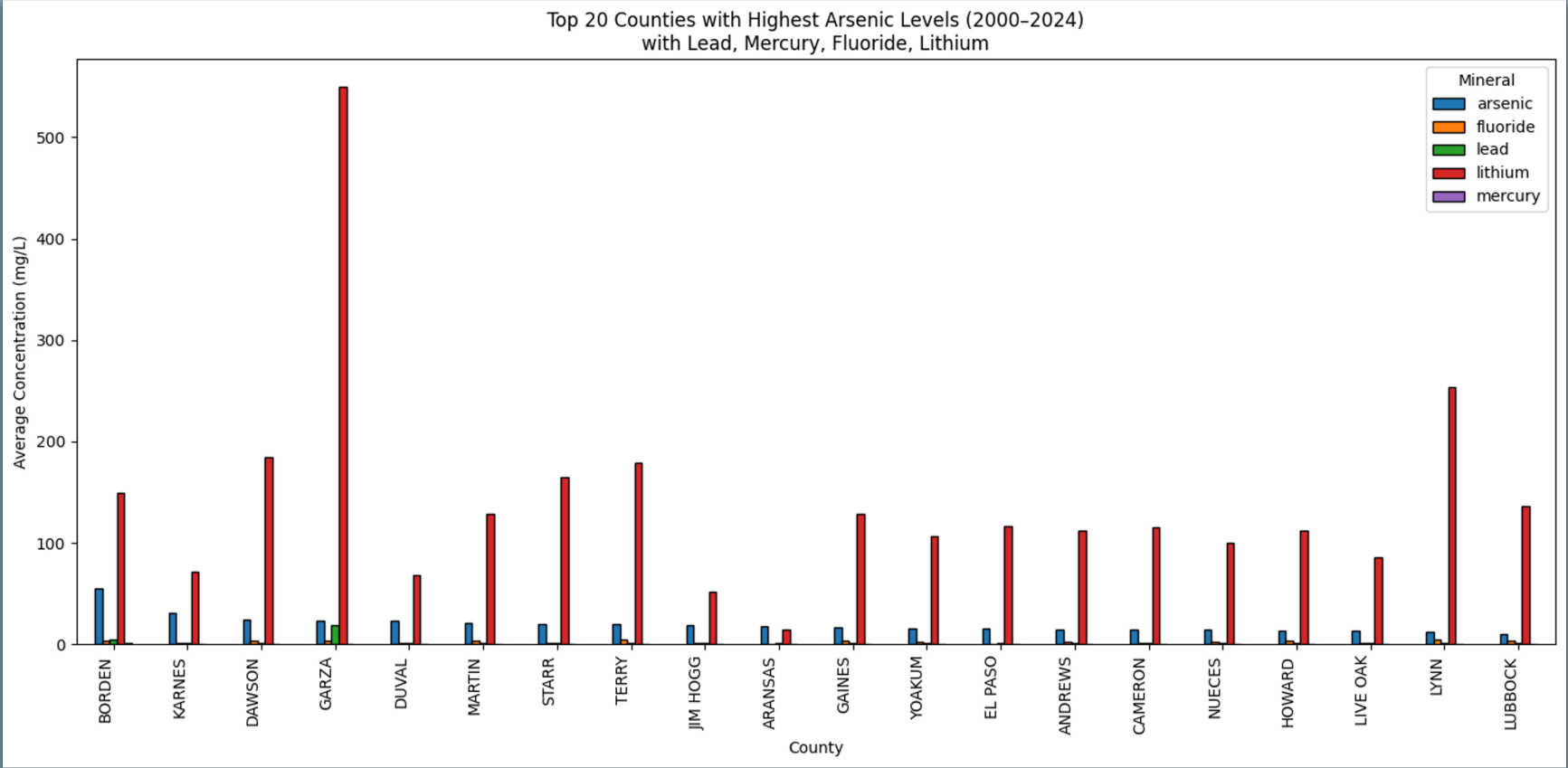
FROM waterqualitydetail wqd
JOIN waterqualityheader wh
    ON wqd.WaterQualityHeaderId = wh.WaterQualityHeaderId
JOIN welldata wd
    ON wh.StateWellId = wd.StateWellId
JOIN welllocation wl
    ON wd.StateWellId = wl.StateWellId
JOIN county c
    ON wl.CountyID = c.CountyID
JOIN waterqualityparameters wqp
    ON wqd.WaterQualityParameterId = wqp.WaterQualityParameterID
WHERE LOWER(wqp.ConstituentName) IN ('arsenic', 'lead', 'mercury', 'fluoride', 'lithium')
    AND wh.SampleDate BETWEEN '2000-01-01' AND '2024-12-31'
GROUP BY c.CountyName
ORDER BY Arsenic DESC;
```

SQL
Output



County	Arsenic	Lea_d	Mercury	Fluoride	Lithium
BORDEN	55.57	5.14	0.83	3.29	149.25
KARNES	31.39	1.16	0.20	0.95	71.41
DAWSON	24.57	1.34	0.67	3.92	184.33
GARZA	23.34	18.56	0.51	3.11	549.59
DUVAL	23.25	1.05	0.20	1.30	67.89
MARTIN	21.07	1.17	0.65	3.58	128.64
STARR	20.19	1.35	0.20	1.31	165.02
TERRY	19.67	1.02	0.54	4.51	179.31
JIM HOGG	18.91	1.00	0.21	1.31	51.88
ARANSAS	17.69	1.00	0.20	0.30	14.97
GAINES	16.40	1.17	0.64	3.34	128.83
YOAKUM	15.55	1.05	0.63	2.79	106.28
EL PASO	15.37	1.21	0.50	0.76	116.78
ANDREWS	14.52	1.13	0.61	2.43	111.62
CAMERON	14.46	1.56	0.20	0.91	115.17
NUECES	14.26	1.26	0.20	2.24	98.96

VISUALIZATION BY JUPYTER NOTEBOOK



COUNT OF SAMPLES PER MINERAL

-- Count of samples per mineral from 2000 to 2024 --

```
SELECT
  LOWER(wqp.ConstituentName) AS Mineral,
  COUNT(*) AS SampleCount
FROM waterqualitydetail wqd
JOIN waterqualityheader wh
  ON wqd.WaterQualityHeaderId = wh.WaterQualityHeaderId
JOIN welldata wd
  ON wh.StateWellId = wd.StateWellId
JOIN welllocation wl
  ON wd.StateWellId = wl.StateWellId
JOIN county c
  ON wl.CountyID = c.CountyID
JOIN waterqualityparameters wqp
  ON wqd.WaterQualityParameterId = wqp.WaterQualityParameterID
WHERE wh.SampleDate BETWEEN '2000-01-01' AND '2024-12-31'
GROUP BY LOWER(wqp.ConstituentName)
ORDER BY SampleCount DESC;
```

SQL
Output

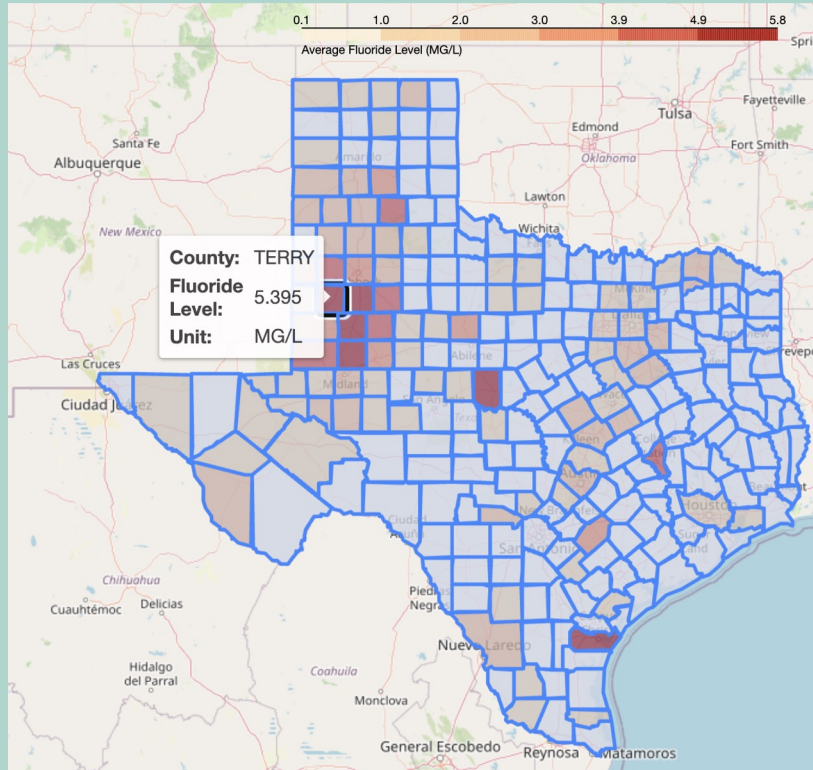
Result Grid			Filter Rows:
	Mineral	SampleCount	
▶	calcium	17735	
	magnesium	17735	
	sodium	17687	
	chloride	17686	
	sulfate	17640	
	nitrate	17535	
	total dissolved solids	17460	
	total hardness	17414	
	fluoride	17357	
	iron	17195	
	potassium	17113	
	arsenic	17076	
	specific conductance	17061	
	manganese	16959	
	zinc	16902	
	copper	16901	
	barium	16883	
	chromium	16859	
	lead	16846	
	selenium	16845	
	cadmium	16820	

PYTHON & MYSQL DEMO

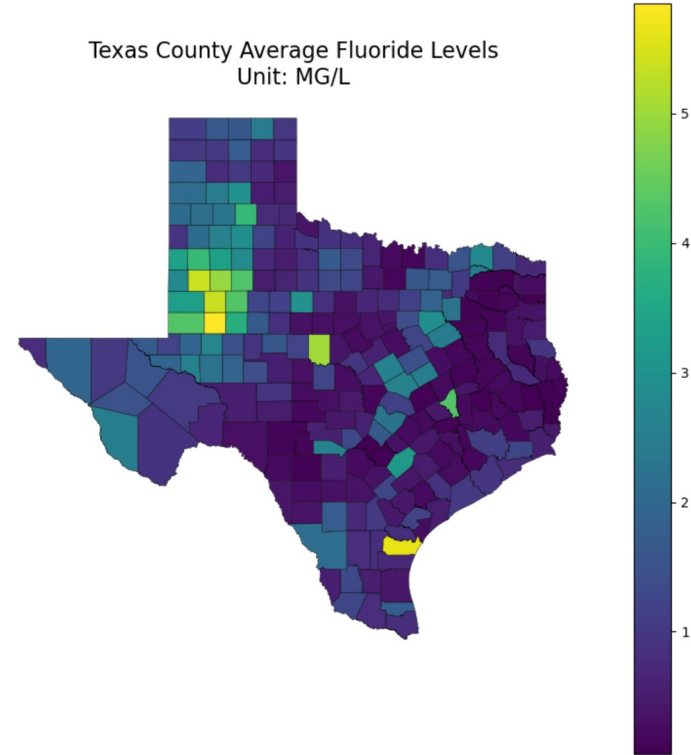
- MySQL Connector/SQL Alchemy SQL queries
- Pandas merges SQL result tables for final data transformation
- GeoPandas + folium for mapping
- **Example:**
 - Load SQL results → merge with Texas county shapefile → generate choropleth

PYTHON DEMO

Fluoride

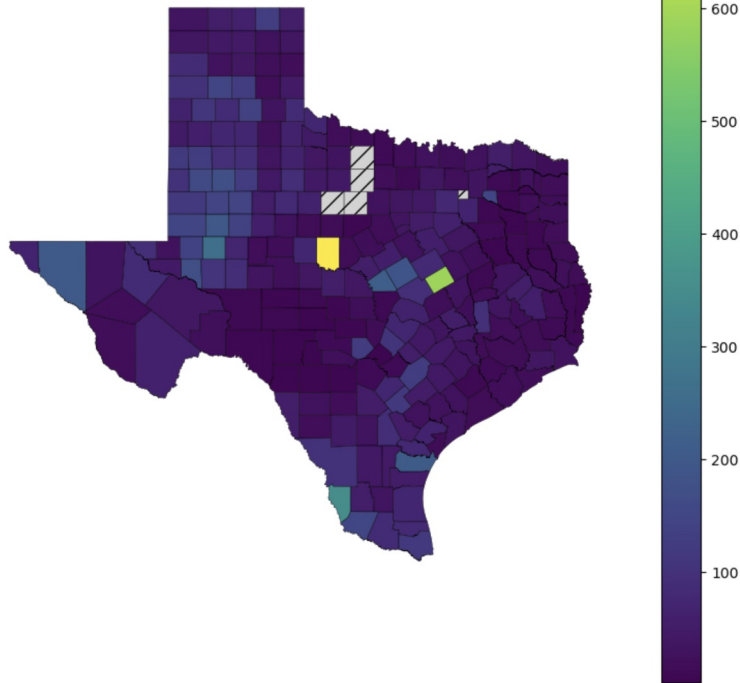


Texas County Average Fluoride Levels
Unit: MG/L

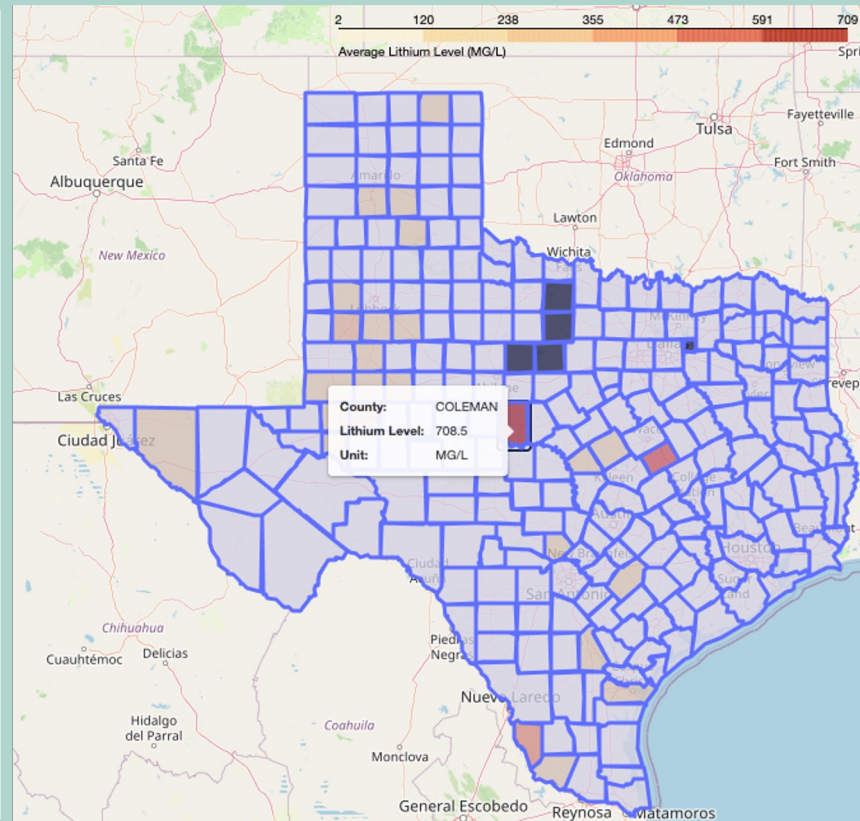


PYTHON DEMO *cont.*

Texas County Average Lithium Levels
Unit: UG/L



Lithium



PRELIMINARY RESULTS

- MySQL database is operational
- SQL joins correctly link wells → samples → minerals → counties
- Identified high- and low-concentration counties for several minerals
- Validated data structure against TWDB documentation
- Successfully pulled data into Python for analysis
- Visualized data

DISCUSSION & LIMITATIONS

STRENGTHS

- Normalized and scalable relational design
- Fast SQL aggregation
- Clean integration with Python
- Supports future ML and geospatial pipelines

LIMITATIONS

- Sparse sampling in rural counties
- Some parameter units inconsistent
- Sampling frequency varies widely
- Not all wells have complete data

FUTURE WORK



- Add health outcome datasets (CDC, NIH, EPA)
- Automate ETL pipeline for new TWDB data and health data
- Time-series trend analysis by county by contaminant and associated outcomes
- Add Python mapping dashboard
- Predictive modeling for contaminant hotspots

Thank you!



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